AC Power Transmission

Power exchange, cables and OHL



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AMERICAN TERAWATT

Abbreviations

AAAC	All-aluminium alloy conductor
AAC	All-aluminium conductor
AC	Alternating current
ACSR	Aluminium conductor steel reinforced
ACSS	Aluminium conductor steel supported
AVR	Automatic voltage regulator
CB	Circuit breaker
DC	Direct current
DS	Disconnector (disconnect switch)
GMD	Geometric mean distance
GMR	Geometric mean radius
HVDC	High-voltage direct current
kcmil	Thousand circular mils
LCC	Line-commutated converter
MOV	Metal-oxide varistor
OHL	Overhead line
PCC	Point of common coupling
PSS	Power system stabiliser
RMS	Root mean square
SCL	Short-circuit level
SCR	Short-circuit ratio
SIL	Surge-impedance loading
SSCI	Sub-synchronous control interaction
SSR	Sub-synchronous resonance
VSC	Voltage-sourced converter
XLPE	Cross-linked polyethylene
pf	Power factor
p.u.	Per unit

Symbols

Sym.	Description
P	Active power
V_{LL}	Line-to-line (phase-phase) voltage
V_{LN}	Line-to-neutral (phase-ground) voltage
δ	Transmission angle between the terminal voltages
X	Series reactance of the path between the terminals
L'	Series inductance per unit length

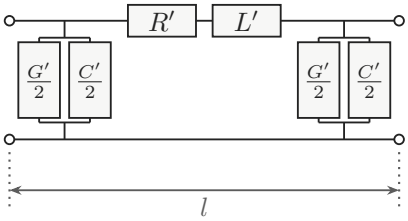
Sym.	Description
D_m	Geometric mean distance between the phases
r_b	Equivalent radius of the conductor bundle
Z_w	Wave (surge) impedance of the line
β	Phase constant, $\beta \cdot l$ the electrical length
K_s	Synchronising power coefficient, $\partial P / \partial \delta$
$\underline{\gamma}$	Propagation constant, $\alpha + j \cdot \beta$
α	Attenuation constant
H	Inertia constant of a synchronous machine

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The line parameters R', L', G' and C'

Per unit length

Four parameters describe the whole line
Two in series, two in shunt — each a distinct physical effect.



	Parameter	Effect	Value
R'	Resistance	Ohmic loss	56.9 mΩ/km
L'	Inductance	Magnetic field	0.891 mH/km
G'	Conductance	Leakage	0.05 µS/km
C'	Capacitance	Electric field	12.7 nF/km

From the same geometry as X: $L' = \frac{\mu_0}{2 \cdot \pi} \cdot \ln \frac{D_m}{r_b}$,
$C' = \frac{2 \cdot \pi \cdot \varepsilon_0}{\ln\left(D_m / r_{b_c}\right)}$. On an OHL
$G' \ll \omega \cdot C' = 4.79\,\mu\text{S/km}$.

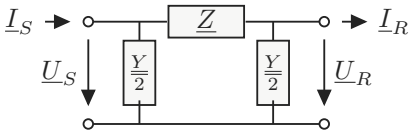
$$\underline{Z}' = R' + j \cdot \omega \cdot L'$$
$$\underline{Y}' = G' + j \cdot \omega \cdot C'$$

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The line as a two-port — the π-section

Where the capacitance enters

The line also charges itself
Series R' and L', shunt C' — a two-port, not one impedance.



$$\underline{Z} = \left(R' + j \cdot \omega \cdot L'\right) \cdot l$$
$$\underline{Y} = \left(G' + j \cdot \omega \cdot C'\right) \cdot l$$

$$\begin{bmatrix} \underline{U}_S \\ \underline{I}_S \end{bmatrix} = \begin{bmatrix} \underline{A} & \underline{B} \\ \underline{C} & \underline{D} \end{bmatrix} \cdot \begin{bmatrix} \underline{U}_R \\ \underline{I}_R \end{bmatrix}$$

$$\underline{A} = \underline{D} = 1 + \frac{\underline{Z} \cdot \underline{Y}}{2}$$

$$\underline{B} = \underline{Z}$$

$$\underline{C} = \underline{Y} \cdot \left(1 + \frac{\underline{Z} \cdot \underline{Y}}{4}\right)$$

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Power transfer of the two-port

The complete relation

Everything the line does sits in $\underline{A}$ and $\underline{B}$
Magnitudes set the amplitude, the angles set the shape.

$$\underline{U}_S = U_S \cdot e^{j\cdot\delta}, \quad \underline{U}_R = U_R, \quad \underline{A} = |A| \cdot e^{j\cdot\vartheta_A}, \quad \underline{B} = |B| \cdot e^{j\cdot\vartheta_B}$$

$$\underline{I}_R = \frac{\underline{U}_S - \underline{A}\cdot\underline{U}_R}{\underline{B}} \qquad \underline{S}_R = \underline{U}_R \cdot \underline{I}_R^*$$

$$P_R = \frac{U_S \cdot U_R}{|B|} \cdot \cos(\vartheta_B - \delta) - \frac{|A| \cdot U_R^2}{|B|} \cdot \cos(\vartheta_B - \vartheta_A)$$

$$Q_R = \frac{U_S \cdot U_R}{|B|} \cdot \sin(\vartheta_B - \delta) - \frac{|A| \cdot U_R^2}{|B|} \cdot \sin(\vartheta_B - \vartheta_A)$$

Short and lossless: $\underline{A} = 1, \underline{B} = j \cdot X \Rightarrow \cos(\vartheta_B - \delta) = \sin\delta$ and the second term vanishes.

$\underline{U}_R$ is the phase reference — δ is the angle by which the sending end leads it.

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AC power transmission — the simplified relation

Principle

Active power flows on the *angle* between two voltages
Reactive power on the *difference in their magnitude*.

$$P = \frac{U_S \cdot U_R}{X} \cdot \sin \delta$$

$$Q_R = \frac{U_S \cdot U_R \cdot \cos \delta - U_R^2}{X}$$

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The series reactance X

Where X comes from

X is set by the geometry of the line
Length enters linearly, the geometry only through a logarithm.

$$X = \omega \cdot L' \cdot l \qquad L' = \frac{\mu_0}{2 \cdot \pi} \cdot \ln \frac{D_m}{r_b}$$

Conductors per phase	r_b / m	X' / Ωkm^{-1}	**rel.** / %
1	0.018	0.499	148
2	0.091	0.377	112
3	0.155	0.336	100
4	0.222	0.310	92

Doubling r_b — or halving D_m — removes only 15 % of X.

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Wave Representation

The line as a distributed system

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From the π-section to the wave equation

Why the lumped model runs out

The parameters are distributed along the line, not concentrated
Two quantities describe the distribution completely.

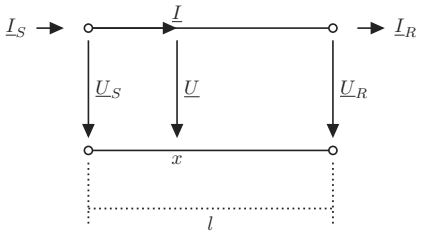
$$\underline{\gamma} = \alpha + j \cdot \beta = \sqrt{\underline{Z'} \cdot \underline{Y'}} = \sqrt{(R' + j \cdot \omega \cdot L') \cdot (G' + j \cdot \omega \cdot C')}$$

$$\underline{Z}_w = \sqrt{\frac{\underline{Z'}}{\underline{Y'}}} = \sqrt{\frac{R' + j \cdot \omega \cdot L'}{G' + j \cdot \omega \cdot C'}}$$

Reference line: $\alpha = 1.07 \times 10^{-4}$ Np/km, $\beta = 1.27 \times 10^{-3}$ rad/km, $\underline{Z}_w = 266 - j\,22\,\Omega$, wavelength 4936 km.

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Voltage and current along the line



At distance x from the sending end

$$\underline{U} = \underline{U}_R \cdot \cosh\left(\underline{\gamma} \cdot (l - x)\right) + \underline{Z}_w \cdot \underline{I}_R \cdot \sinh\left(\underline{\gamma} \cdot (l - x)\right)$$

$$\underline{I} = \frac{\underline{U}_R}{\underline{Z}_w} \cdot \sinh\left(\underline{\gamma} \cdot (l - x)\right) + \underline{I}_R \cdot \cosh\left(\underline{\gamma} \cdot (l - x)\right)$$

At the sending end $(x = 0)$

$$\underline{U}_S = \underline{U}_R \cdot \cosh\left(\underline{\gamma} \cdot l\right) + \underline{Z}_w \cdot \underline{I}_R \cdot \sinh\left(\underline{\gamma} \cdot l\right)$$

$$\underline{I}_S = \frac{\underline{U}_R}{\underline{Z}_w} \cdot \sinh\left(\underline{\gamma} \cdot l\right) + \underline{I}_R \cdot \cosh\left(\underline{\gamma} \cdot l\right)$$

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The exact two-port parameters

Closing the loop

The same $\underline{A}$, $\underline{B}$, $\underline{C}$, $\underline{D}$ as before — now exact
The nominal π was their short-line approximation.

$$\underline{A} = \underline{D} = \cosh\left(\underline{\gamma} \cdot l\right), \qquad \underline{B} = \underline{Z}_w \cdot \sinh\left(\underline{\gamma} \cdot l\right), \qquad \underline{C} = \frac{\sinh\left(\underline{\gamma} \cdot l\right)}{\underline{Z}_w}$$

l	$\beta \cdot l$	$\lvert B \rvert$ **nominal** π	$\lvert B \rvert$ **exact**	**error**
mi	°	Ω	Ω	%
100	11.7	54.8	54.5	$+0.7$
200	23.5	109.7	106.7	$+2.8$
300	35.2	164.5	154.4	$+6.5$
400	46.9	219.4	195.8	$+12.0$
600	70.4	329.1	252.9	$+30.1$

The π-section holds to roughly 100 mi; beyond that only the wave form is usable.

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The 500 kV reference line

Worked example

The same line carries the whole chapter
The model you choose moves P_{max} by up to 20 %.

Line		Value
U_n	kV	500
f	Hz	60
l	mi (km)	300 (483)
Conductor		$3\times$ Bluebird
Geometry	ft / in	35 flat / 18
R'	mΩ/km	56.9
X'	Ω/km	0.336
C'	nF/km	12.7
Z_w	Ω	266
P_{nat}	MW	939

At 300 mi		Value
$\underline{Z}$	Ω	$27.5 + j\,162.2$
$\underline{Y}$	mS	$j\,2.31$
Charging at 1.0 p.u.	MVAr	578
$\beta \cdot l$	°	35.2

Model	$\lvert A\rvert$	ϑ_B °	$\lvert B\rvert$ Ω	P_{max} MW
Short, lossless	1.000	90.0	162.2	1541
Nominal π	0.813	80.4	164.5	1266
Distributed	0.819	81.0	154.4	1364

P_{max} occurs at $\delta = \vartheta_B$; the simplification moves it to 90° and overstates it by 13 %.

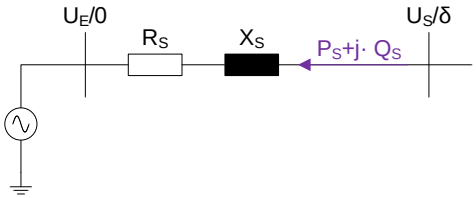
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Power transmission

Principle definition

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Power exchange in the AC grid



P_S is the export active power of the VSC converter station at PCC

Q_S is the export reactive power of the VSC converter station at PCC

U_S is the RMS value of the fundamental frequency component of the line-to-line AC voltage of the VSC converter station at PCC

U_E is the equivalent AC voltage of the AC grid, corresponding to a SCL at PCC

R_S is the equivalent resistance of the SCL impedance at PCC

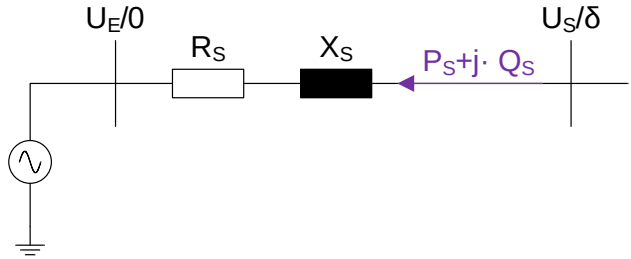
X_S is the equivalent reactance of the SCL impedance at PCC

δ is the phase angle difference between AC voltages U_S and U_E

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Power exchange in the AC grid



$$P_S = \frac{U_S}{R_S^2+X_S^2} \cdot [(U_S - U_E \cdot \cos\delta) \cdot R_S + U_E \cdot X_S \cdot \sin\delta]$$

$$Q_S = \frac{U_S}{R_S^2+X_S^2} \cdot [(U_S - U_E \cdot \cos\delta) \cdot X_S - U_E \cdot R_S \cdot \sin\delta]$$

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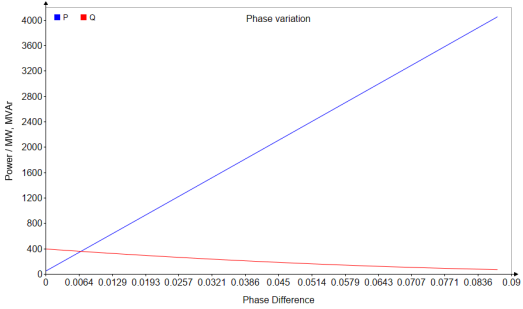
Power exchange

Parameter	Case 1	Case 2	Case 3	Case 4
P_S	1000 MW	1500 MW	1000 MW	-1000 MW
U_E	230 kV	230 kV	230 kV	230 kV
U_S	232 kV	232 kV	240 kV	232 kV
R_S	0.144 Ω	0.144 Ω	0.144 Ω	0.144 Ω
X_S	1.15 Ω	1.15 Ω	1.15 Ω	1.15 Ω
δ	1.19°	1.816°	1.081°	$-1.319°$
Q_S	288 MVAr	239 MVAr	905 MVAr	541 MVAr

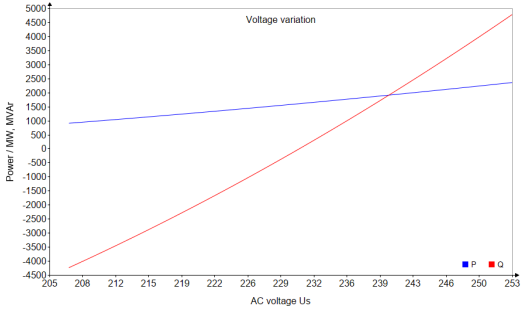
- Active power transmission requires reactive power.
- Higher AC voltage increases the reactive-power demand.
- Changing the power-flow direction also changes the reactive-power demand.

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Power exchange — active vs. reactive



Active power vs. voltage difference

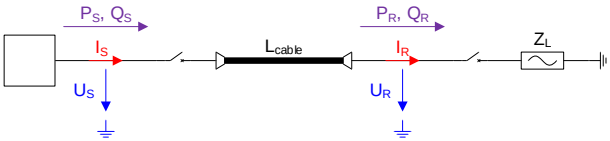


Active power vs. phase difference

Active power is transmitted via phase difference; reactive power via voltage amplitude.

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Power transmission in the AC grid



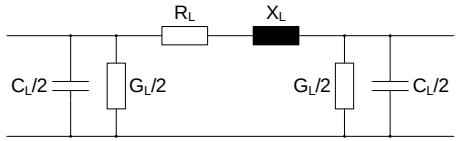
- AC power transmission principles shown with a simple AC network.
- The grid is connected to the load with impedance Z_L via an AC cable of length L_{cable}.

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Transmission line as 2-port units

Param.		OHL	Cable (XLPE)
C'_L	nF/km	10	180–420
R'_L	Ω/km	0.1–1	0.1–1
X'_L	mH/km	1	0.31–1.42
G'_L	µS/km	0.05	1

- AC line represented by a π-section model.
- Cables are dominated by capacitance.
- OHL are dominated by inductance.



Example cable parameters:

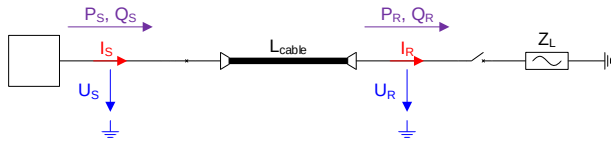
$C'_L = 190\,\text{nF/km}$, $R'_L = 22\,\text{m}\Omega\text{/km}$,

$X'_L = 1.35\,\text{mH/km}$

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Power transmission — no load connected

Param.	Value
$U_{S,LL}$	230 kV (1.0 p.u.)
I_S	1060 A
P_S	2.2 MW
Q_S	-421 MVAr
$U_{R,LL}$	1.22 p.u.
I_R	0 A
L_{cable}	60 mi

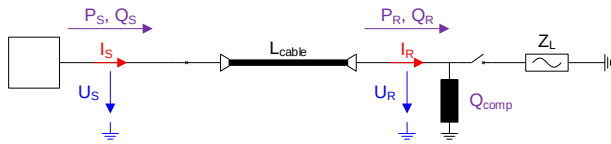


Connecting an AC cable creates significant current flow and reactive-power demand.

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Power transmission — reactive compensation

Param.	Value
$U_{S,LL}$	230 kV (1.0 p.u.)
I_S	40 A
P_S	2.2 MW
Q_S	-14.4 MVAr
I_R	830 A
Q_R	-277 MVAr
Q_{comp}	400 MVAr

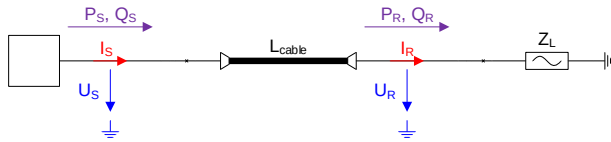


An uncompensated cable connection is usually not allowed; a shunt reactor Q_{comp} is required.

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Power transmission — with load

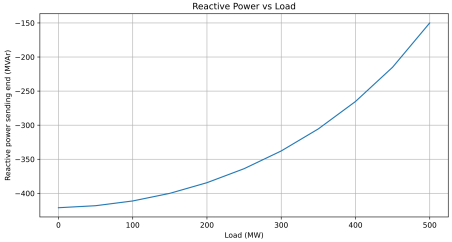
Param.	Value
$U_{S,LL}$	230 kV (1.0 p.u.)
I_S	1340 A
P_S	511 MW
Q_S	-150 MVAr
$U_{R,LL}$	1.06 p.u.
I_R	1180 A
P_R	500 MW



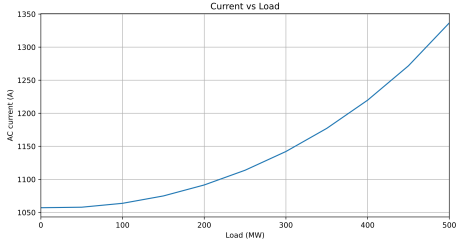
Power transmission reduces the voltage rise and the reactive-power demand.

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Power transmission — reactive power & current vs. active power



Reactive power vs. active power



AC current vs. active power

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Natural Power

Surge-impedance loading of an AC line

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Natural power and surge-impedance loading

A line terminated in its own characteristic impedance $\underline{Z}_w$ carries its *natural power*: the reactive power generated by the shunt capacitance exactly cancels the reactive power consumed by the series inductance, so the line exchanges no reactive power and its voltage magnitude stays constant. North American practice calls the same quantity the *surge-impedance loading* (SIL).

$$\underline{Z}_w = \sqrt{\frac{\underline{Z}'}{\underline{Y}'}} = \sqrt{\frac{R' + j \cdot \omega \cdot L'}{G' + j \cdot \omega \cdot C'}} \approx \sqrt{\frac{L'}{C'}} = Z_w \qquad P_{nat} = \mathsf{SIL} = \frac{U^2}{Z_w}$$

- On an overhead line $\omega \cdot L' \gg R'$ and $\omega \cdot C' \gg G'$, so the real $\sqrt{L'/C'}$ is within a percent of $|\underline{Z}_w|$ — the value P_{nat} is quoted from.
- Z_w is pure conductor geometry: independent of line length, and of frequency in the lossless form.
- P_{nat} is a reference, not a rating: below it the line sources reactive power, above it absorbs it.

U line-to-line voltage; R', L', G', C' per-unit-length resistance, inductance, shunt conductance and capacitance; $\omega = 2 \cdot \pi \cdot f$.

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Natural power — definition

Voltage and current at a distance x from the receiving end follow from the two-port equations of the uniform line:

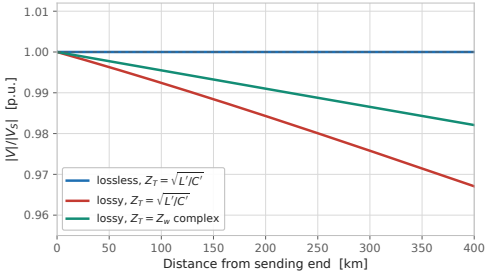
$$\underline{U}(x) = \underline{U}_R \cdot \cosh\left(\underline{\gamma} \cdot x\right) + \underline{Z}_w \cdot \underline{I}_R \cdot \sinh\left(\underline{\gamma} \cdot x\right)$$

$$\underline{I}(x) = \frac{\underline{U}_R}{\underline{Z}_w} \cdot \sinh\left(\underline{\gamma} \cdot x\right) + \underline{I}_R \cdot \cosh\left(\underline{\gamma} \cdot x\right)$$

Closing the line with $\underline{Z}_w$ makes $\underline{I}_R = \underline{U}_R / \underline{Z}_w$, and both collapse to a single travelling wave — no reflection:

$$\underline{U}(x) = \underline{U}_R \cdot e^{\underline{\gamma} \cdot x}, \quad \underline{I}(x) = \underline{I}_R \cdot e^{\underline{\gamma} \cdot x}$$

The magnitude then falls only by the attenuation $e^{-\alpha \cdot l}$, and $\underline{U}/\underline{I} = \underline{Z}_w$ everywhere.



Matched line, 400 km — only the complex $\underline{Z}_w$ matches perfectly.

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Natural power — the reactive balance

The same result follows from the reactive power balance of a line of length l at line-to-line voltage U and phase current I:

$$Q_C = \omega \cdot C' \cdot l \cdot U^2 \qquad \text{(shunt, generated)}$$

$$Q_L = 3 \cdot \omega \cdot L' \cdot l \cdot I^2 \qquad \text{(series, consumed)}$$

Natural loading is the point at which the two cancel. Setting $Q_C = Q_L$ eliminates l and ω and gives $I = U/\left(\sqrt{3} \cdot Z_w\right)$, hence

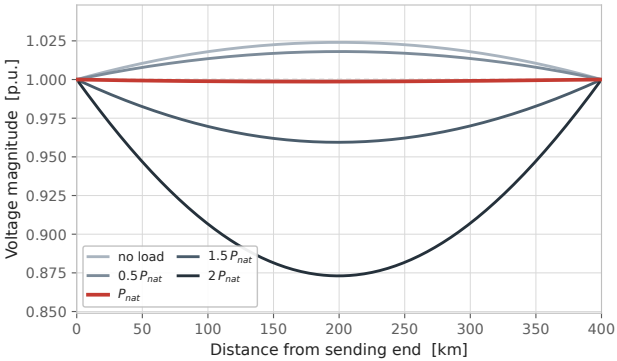
$$\boxed{P_{nat} = \sqrt{3} \cdot U \cdot I = \frac{U^2}{\sqrt{L'/C'}} = \frac{U^2}{Z_w}}$$

Quantity		Value
R'	mΩ/km	25
X'	Ω/km	0.30
$L' = X'/\omega$	mH/km	0.955
C'	nF/km	12.5
$Z_w = \sqrt{L'/C'}$	Ω	276.4
$\lvert\underline{Z}_w\rvert$	Ω	276.9
$\beta \cdot l$	°	24.88
P_{nat}	MW	579

400 kV, 2× bundle, 50 Hz, $l = 400$ km, $G' = 0$.

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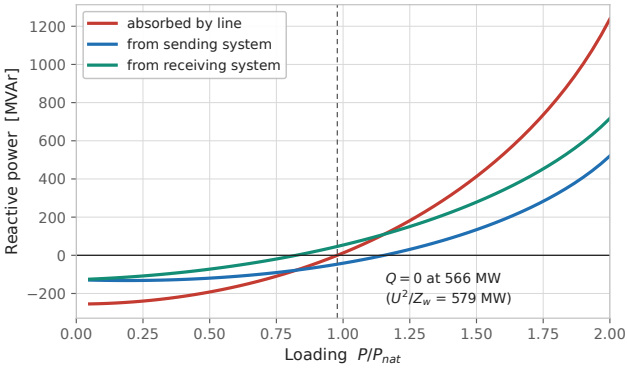
Voltage profile along the line



- Both terminals held at 1.0 p.u.; the transmission angle ϑ is solved for each loading.
- Below P_{nat} the shunt capacitance dominates and the middle of the line rises — 1.024 p.u. at no load.
- At P_{nat} the profile is flat to four decimals.
- Above P_{nat} the series reactance dominates and the middle sags — 0.873 p.u. at $2 \cdot P_{nat}$, with $\vartheta = 63°$ across the line.

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Reactive balance and the effect of the line losses



The net reactive power absorbed by the line follows from the terminal powers, $Q_{line} = \Im\{\underline{S}_S\} - \Im\{\underline{S}_R\}$.

- Lossless line: the zero crossing reproduces U^2/Z_w to six decimals.
- Real line: 566 MW, i.e. 2.2 % below the 579 MW of U^2/Z_w.
- Cause — the sending-end current carries the load *plus* the losses, so $3 \cdot I^2 \cdot X' \cdot l$ is larger and the balance is reached at a lower transfer.

Two-port solution; verified against a 20-segment pandapower model (deviation $< 0.1\,\%$).

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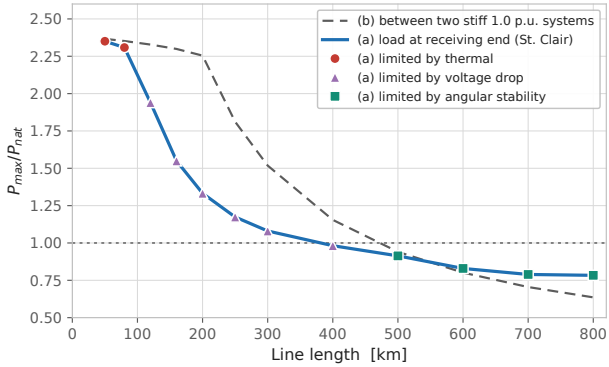
Natural power of typical line types

Line	U	C'	Z_w	P_{nat}	S_{th}/P_{nat}
	kV	nF/km	Ω	MW	
230 kV, 1× Drake	230	9.0	376.1	141	2.8
345 kV, 2× bundle	345	11.8	292.3	407	2.6
400 kV, 2× bundle	400	12.5	276.4	579	2.4
500 kV, 3× bundle	500	13.5	256.6	974	2.3
765 kV, 4× bundle	765	13.5	250.8	2334	2.0

- Z_w spans only 251–376 Ω while P_{nat} spans 141–2334 MW: P_{nat} is a U^2 quantity.
- Bundling and tighter phase spacing are the only levers on Z_w — both lower L' and raise C'.
- S_{th}/P_{nat} decides where the thermal plateau of the loadability curve ends.
- AC cables sit at $Z_w \approx 20 \ldots 40\,\Omega$: their P_{nat} is far above the thermal rating, so charging current is the binding constraint instead.

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Loadability — natural power is a reference, not a limit



Criteria: $I \leq 2.0\,\text{kA}$, $U_R \geq 0.95\,\text{p.u.}$, $\vartheta \leq 30°$, no compensation.

(a) load at receiving end

≤ 80 km	thermal, $2.3 \ldots 2.4 \cdot P_{nat}$
120–400 km	voltage drop
≥ 500 km	angular stability
382 km	falls through $1 \cdot P_{nat}$

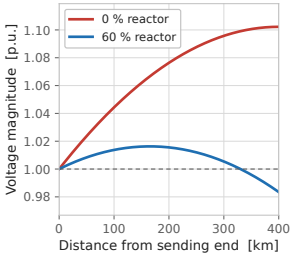
(b) two stiff systems

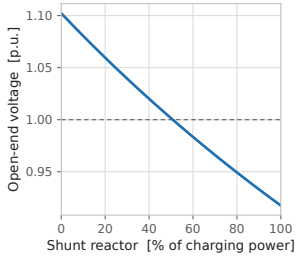
≤ 160 km	thermal
≥ 200 km	angular stability
472 km	falls through $1 \cdot P_{nat}$

The boundary condition matters: holding U_R at 1.0 p.u. (case b) removes the voltage-drop criterion entirely and shifts the $1 \cdot P_{nat}$ crossing by 90 km. Case (a) is the classical St. Clair configuration.

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Light load — the Ferranti effect





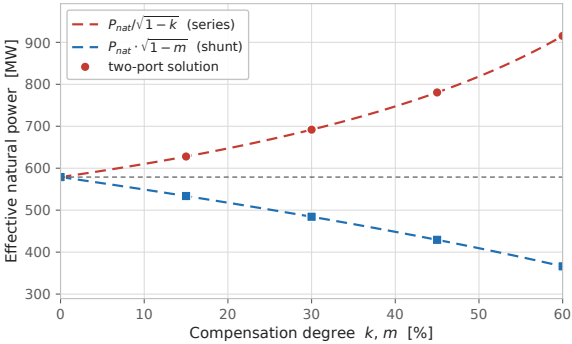
Open-ended line, $\underline{I}_R = 0$:

$$\frac{\underline{U}_R}{\underline{U}_S} = \frac{1}{\cosh(\underline{\gamma} \cdot l)} \longrightarrow \frac{1}{\cos(\beta \cdot l)}$$

- $\beta \cdot l = 24.88°$ gives 1.102 p.u. at the open end, from 251 MVAr of charging power.
- A 60 % shunt reactor brings it back to 0.984 p.u.; full compensation gives 0.918 p.u.
- The degree follows from the light-load voltage, but it moves P_{nat} too.

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Compensation moves the natural power



For *uniformly distributed* compensation the scaling is exact:

$$X' \to X' \cdot (1-k) \quad \Rightarrow \quad \frac{P_{nat}}{\sqrt{1-k}}$$

$$C' \to C' \cdot (1-m) \quad \Rightarrow \quad P_{nat} \cdot \sqrt{1-m}$$

- $k = 60$ % raises P_{nat} by $1.581\times$; $m = 60$ % lowers it by $0.632\times$.
- Real banks are lumped at substations, so the laws are a first-order guide only: a lumped 30 % series bank at $1.4 \cdot P_{nat}$ reduces ϑ from $37.6°$ to $25.6°$ and turns the sending-end reactive flow from $+89$ to -30 MVAr.

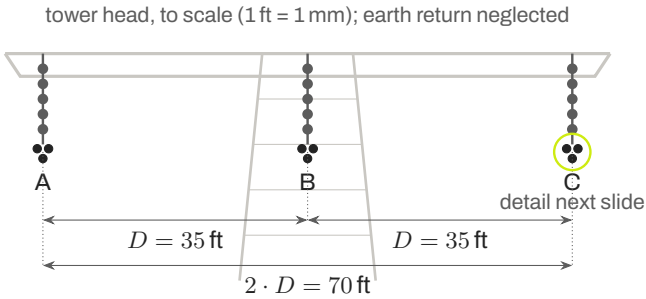
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OHL Configuration

Geometry of the 500 kV reference line

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Tower head — flat configuration



Parameter		Value
D phase spacing	ft	35
D_{AC} outer phases	ft	70
Conductors per phase		3
s bundle spacing	in	18
d conductor diameter	in	1.762
GMR conductor	ft	0.0588

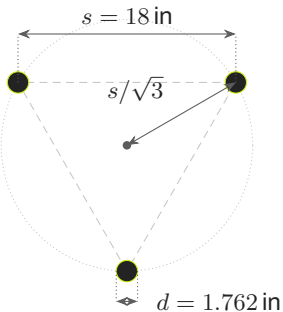
Bluebird ACSR, 2156 kcmil, 84/19, single circuit.

- Flat configuration: lowest towers, widest right-of-way; the outer phases see $2 \cdot D$.
- GMD on the next slide assumes a transposed line.

Conductor data: Southwire ACSR bare conductor tables, catalog section 11-4. Tower body and insulator lengths are schematic.

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Bundle detail and equivalent spacings



bundle, to scale (1 in = 1.6 mm)

$$GMR_b = \sqrt[3]{GMR \cdot s^2} \qquad \text{(bundle, } L'\text{)}$$

$$r_b = \sqrt[3]{\frac{d}{2} \cdot s^2} \qquad \text{(bundle, } C'\text{)}$$

$$GMD = \sqrt[3]{D \cdot D \cdot 2 \cdot D} = D \cdot \sqrt[3]{2} \qquad \text{(phases)}$$

$$L' = \frac{\mu_0}{2 \cdot \pi} \cdot \ln \frac{GMD}{GMR_b}, \qquad C' = \frac{2 \cdot \pi \cdot \varepsilon_0}{\ln\left(GMD/r_b\right)}$$

Quantity		$3\times$	$2\times$
GMR_b	in	6.11	3.56
r_b	in	6.58	3.98
GMD	ft	44.1	44.1
L'	mH/km	0.892	1.000
C'	nF/km	12.68	11.38
$Z_w = \sqrt{L'/C'}$	Ω	265	296

For a two-conductor bundle $GMR_b = \sqrt{GMR \cdot s}$, $r_b = \sqrt{d/2 \cdot s}$. The third subconductor lowers Z_w by ≈ 10 % and so raises P_{nat} by ≈ 11 %.

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Power Transmission Limits

Thermal, voltage and stability limits of an OHL

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Three limits on the power transfer of a line

- **Thermal.** Conductor temperature sets the current; independent of length. Binding only on short lines.
- **Voltage drop.** The receiving-end voltage must stay in its band under load: $U_R \geq 0.95$ p.u. with $U_S = 1.0$ p.u.
- **Steady-state stability.** δ is kept well below $90°$ so the system survives the loss of a parallel element; $\delta \leq 30°$ leaves a factor 2 on the lossless limit.
- The *loadability* is the minimum of the three; against length it is the St. Clair curve.

$$P_{th} = \sqrt{3} \cdot U_n \cdot I_{th}$$

$$U_S = |A \cdot \underline{U}_R + B \cdot \underline{I}_R| \leq 1.0\,\text{p.u.}$$

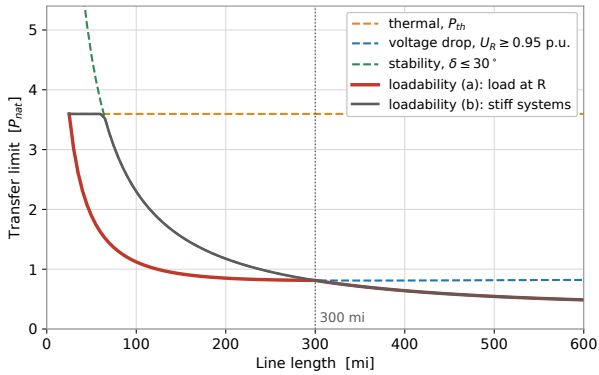
$$P_{stab} \approx \frac{U^2}{(1-k) \cdot X_L} \cdot \sin \delta_{max}$$

Reference line		Value
I_{th}, 3× Bluebird at 75 °C	A	3900
P_{th}	MW	3377
P_{nat}	MW	939

A, B two-port constants of the (compensated) line. The curves use the exact two-port power–angle relation for P_{stab}; the lossless form is shown for orientation.

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Loadability of the 500 kV reference line

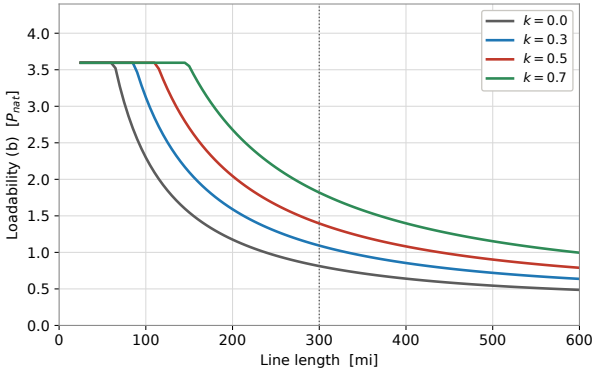


- **(a)** load at the receiving end, no reactive support: thermal to 25 mi, then voltage drop; falls through $1 \cdot P_{nat}$ at 125 mi.
- **(b)** two stiff systems, both ends at 1.0 p.u.: thermal to 60 mi, then angular stability; falls through $1 \cdot P_{nat}$ at 240 mi.
- At 300 mi both cases give 763 MW $(0.81 \cdot P_{nat})$ — the voltage and the angle criterion happen to coincide; the line cannot carry its natural power without compensation.

Uncompensated line. Voltage criterion with a unity power-factor load; a lagging load lowers it further, reactive support at the receiving end raises it.

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Loadability with series compensation



k	P_{stab} MW	P_U MW	(b) P_{nat}	thermal to mi
0	763	763	0.81	60
0.3	1027	826	1.09	85
0.5	1311	844	1.40	110
0.7	1708	796	1.82	145

- Case (b): the bank acts on the stability limit — $k = 0.5$ lifts the 300 mi loadability from 0.81 to $1.40 \cdot P_{nat}$.
- P_U of case (a) barely moves: the $I \cdot X_C$ rise is in quadrature with a unity-pf load, so reactive support at the receiving end is needed as well.

Values at 300 mi, midpoint bank. Curves show case (b), the planning case for a line between two strong systems.

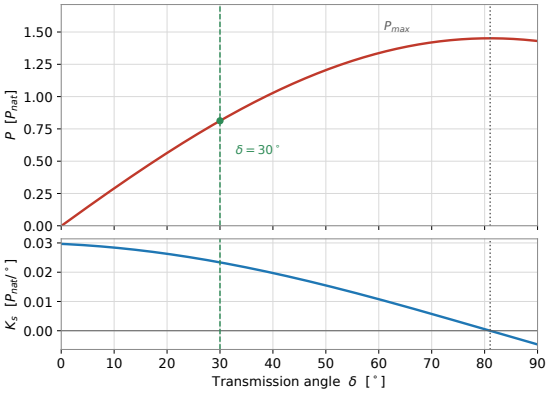
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Angular Stability

Why the transmission angle is held far below $90°$

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The power–angle relation and the synchronising power



$$P = \frac{U_S \cdot U_R}{X} \cdot \sin\delta$$

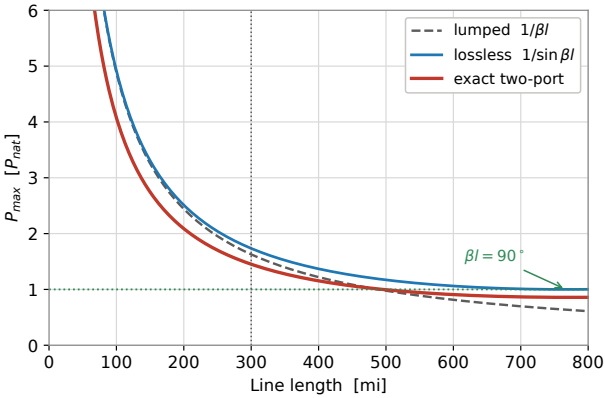
$$K_s = \frac{\partial P}{\partial \delta} = \frac{U_S \cdot U_R}{X} \cdot \cos\delta$$

- K_s is the stiffness of the coupling. Below P_{max} a machine that advances delivers more power and is pulled back; beyond it, less — and it pulls out.
- $K_s = 0$ at P_{max} is the steady-state stability limit. With losses it sits at 81°, not 90°.
- At $\delta = 30^\circ$: $0.81 \cdot P_{nat}$, at 79 % of the stiffness at $\delta = 0$.

Exact two-port, both terminals at 1.0 p.u., 300 mi reference line; the lossless forms above are shown for orientation.

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Electrical length, not lumped reactance



$$X' = Z_w \cdot \sin(\beta \cdot l)$$

$$P_{max} = \frac{U^2}{X'} = \frac{P_{nat}}{\sin(\beta \cdot l)}$$

- $\beta = \omega \cdot \sqrt{L' \cdot C'}$: at 60 Hz the wavelength is 4937 km, a quarter wave 767 mi (1234 km).
- At a quarter wave $P_{max} = P_{nat}$ exactly — the analytical spine under the St. Clair curve.
- 300 mi: lumped $\omega \cdot L' \cdot l = 162.4\,\Omega$ against $X' = 153.4\,\Omega$, and $1/\sin(\beta \cdot l) = 1.73$ against 1.45 exact.

Lumped X overstates beyond roughly 250 km; $R = 27.5\,\Omega$ and the line charging account for the rest of the gap at 300 mi.

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The reactance is not only the line

- The X of the power–angle relation is the whole path between the two internal EMFs, not the line alone.
- A 300 mi line into systems of 10 GVA at each end loses a quarter of its P_{max} to the terminals.
- Loadability quoted for a line in isolation is therefore optimistic; the short-circuit level at both ends belongs in the statement.
- The same quantity drives converter performance, where it is normalised as the SCR.

$$X_{tot} = X_{th_s} + X_{tr_s} + X' + X_{tr_r} + X_{th_r}$$

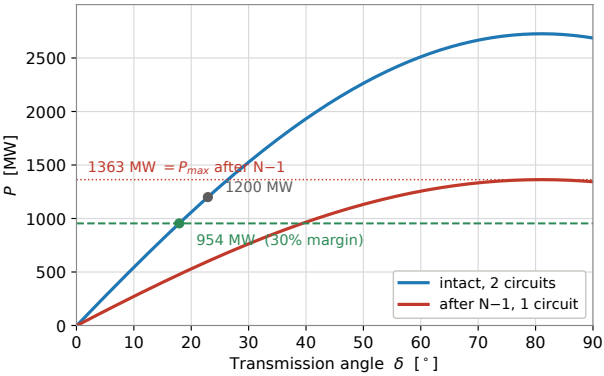
$$\mathrm{SCR} = \frac{S_{sc}}{P_n}, \qquad X_{th} = \frac{U_n^2}{S_{sc}}$$

SCL, each end GVA	X_{th} Ω	X_{tot} Ω	P_{max} MW
5	50.0	253.4	986
10	25.0	203.4	1229
20	12.5	178.4	1401
stiff	0.0	153.4	1630

Lossless U^2/X_{tot}, transformers neglected, 300 mi reference line with $X' = 153.4\,\Omega$. S_{sc} short-circuit level, P_n rated power of the infeed.

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From the maximum to the operating limit

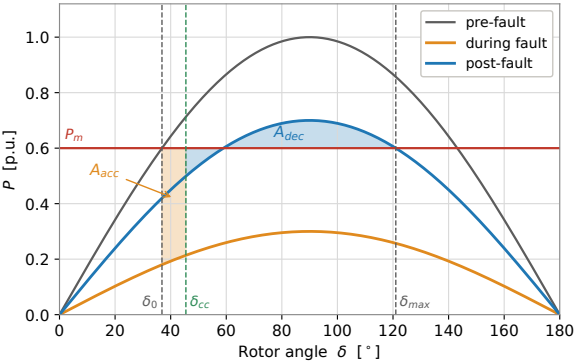


- Two 300 mi circuits: 2726 MW intact, 1363 MW on the survivor after N−1. The reactance of the path doubles.
- Planning criterion $(P_{max} - P)\,/P_{max} \geq 0.30$ on the *post-contingency* network $\rightarrow$ 954 MW, i.e. $1.02 \cdot P_{nat}$ for the pair.
- 1200 MW looks comfortable intact at $\delta = 22.9^{\circ}$ and sits 12 % below the post-contingency maximum — not acceptable.
- The thermal rating of the pair is 6755 MW: seven times the binding constraint.

Both terminals at 1.0 p.u., uncompensated, exact two-port. Series compensation raises the post-contingency curve — see the next section.

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Transient stability — the equal-area criterion



$$A_{acc} = \int_{\delta_0}^{\delta_{cc}} (P_m - P_{flt})\,\mathrm{d}\delta$$

$$A_{dec} = \int_{\delta_{cc}}^{\delta_{max}} (P_{pst} - P_m)\,\mathrm{d}\delta$$

- The output collapses during the fault while the mechanical input continues; the rotor must repay that energy before δ_{max}.
- A higher pre-fault δ shrinks A_{dec} from both ends, so the critical clearing time falls steeply with loading.
- Fast clearing — two-cycle breakers, independent-pole tripping — buys capability.

Generic machine p.u.: $P_m = 0.6$, P_{max} 1.0 pre-fault, 0.3 during the fault, 0.7 after it. $\delta_0 = 36.9°$, $\delta_{cc} = 45.5°$, $\delta_{max} = 121.0°$.

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Damping sets the limit before P_{max} does

- As δ grows K_s falls: the electromechanical mode slows and its damping ratio drops.
- High-gain AVRs — what allows a large δ in the first place — give negative damping torque at heavy loading. The PSS exists to put it back.
- In a multi-machine system the limit is the eigenvalue criterion, not P_{max}: it can bind well below the two-machine maximum.

$$\frac{2 \cdot H}{\omega_0} \cdot \Delta\ddot{\delta} + K_D \cdot \Delta\dot{\delta} + K_s \cdot \Delta\delta = 0$$

$$\omega_n = \sqrt{\frac{K_s \cdot \omega_0}{2 \cdot H}}$$

Mode	Hz
Inter-area	0.2–0.8
Local plant	0.8–2.0
Torsional, SSR/SSCI	10–50
Criterion: damping ratio $\geq 3 \ldots 5$ %	

H inertia constant, K_D damping torque coefficient, ω_0 rated angular frequency. Inter-area modes are the ones that constrain long transfers.

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A DC link has no transmission angle

- No angle, no synchronising torque, no equal-area criterion: the converter control sets power as a commanded quantity.
- Length enters only through the conductor resistance, so the $1/\sin(\beta \cdot l)$ decay has no counterpart.
- An embedded HVDC link can be modulated to *damp* the inter-area modes of the surrounding AC system instead of loading them.
- The constraint moves to the terminals: SCR and commutation robustness for LCC, control interaction for VSC.

	AC line	**HVDC link**
Power set by	δ across X	converter control
Length	$1/\sin \beta l$	R only
Synchronising	inherent	by control
Binding limit	stability	AC strength, converter rating

The angular limits of the preceding slides are the reason a long AC corridor needs compensation and a DC corridor does not.

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Series Compensation

Shortening the electrical length of an OHL

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Series compensation — principle

A capacitor bank in series with the line cancels part of the series inductive reactance. The *degree of compensation* k is defined against the total series reactance of the line:

$$k = \frac{X_C}{X_L}, \qquad X_L = \omega \cdot L' \cdot l, \qquad X_{eff} = (1-k) \cdot X_L$$

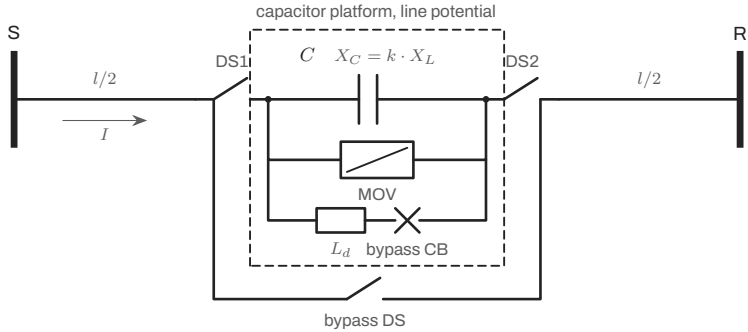
$$P = \frac{U^2}{(1-k) \cdot X_L} \cdot \sin\delta \quad \Rightarrow \quad P_{max} \approx \frac{U^2}{(1-k) \cdot X_L} \qquad \text{(lossless, } |U_S| = |U_R| = U\text{)}$$

- The line behaves as if it were shorter: for a given power the transmission angle δ falls and the steady-state stability margin rises.
- The rise $I \cdot X_C$ across the bank scales with the current: self-regulating, unlike shunt compensation.
- Practical range $k \approx 0.25 \ldots 0.70$; above that SSR, distance-protection behaviour and bank cost dominate.

X_C reactance of the bank at ω; l line length; δ angle between sending- and receiving-end voltages.

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Series compensation — single-line diagram



- MOV limits the capacitor voltage under fault current; the bypass CB with damping reactor L_d takes over at the MOV energy limit.
- DS1/DS2 isolate the platform; the bypass DS keeps the line in service with the bank out ($k = 0$).

One phase shown; one platform per phase at the line midpoint of the reference case. Symbols IEC 60617.

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Series compensation — model and cases

The two-port model of the natural-power slides is reused; the bank is cascaded at the midpoint as a series element $\begin{bmatrix} 1 & -j \cdot X_C \\ 0 & 1 \end{bmatrix}$ between the two half-lines.

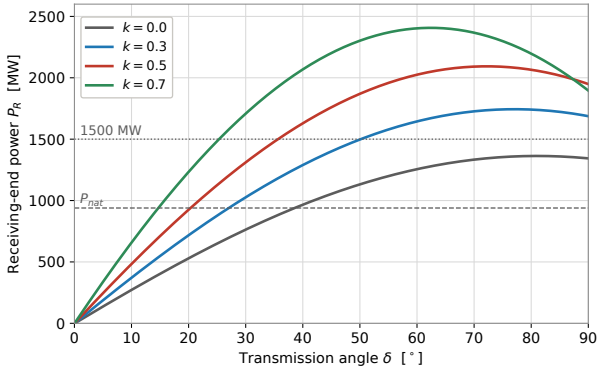
- $\beta \cdot l = 35.2°$: at 300 mi the line is well beyond the thermal plateau of the loadability curve.
- The compensation degree is swept over $k = 0,\ 0.3,\ 0.5,\ 0.7$.

Quantity		Value
X_L	Ω	162.4
R	Ω	27.5
Z_w	Ω	266
P_{nat}	MW	939

The line and its geometry are given on the reference-line slide; only the quantities the compensation uses are repeated here.

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Power–angle curves



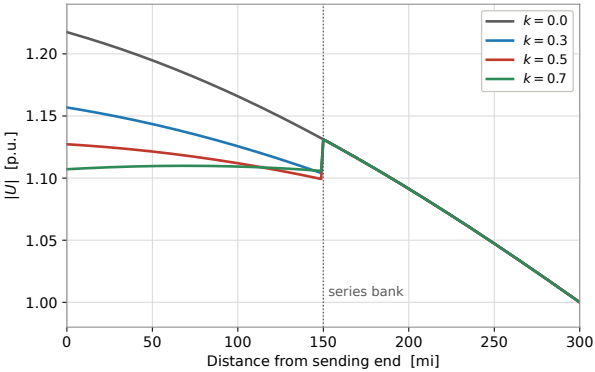
k	X_C Ω	P_{max} MW	δ **at 1500 MW** °
0	0	1363	—
0.3	48.7	1744	50.1
0.5	81.2	2093	35.7
0.7	113.7	2407	25.3

- Uncompensated, 1500 MW ($1.6 \cdot P_{nat}$) cannot be carried at all; $k = 0.5$ carries it at $36°$.
- P_{max} moves to smaller angles as k grows — the line gets electrically shorter.

Both terminals at 1.0 p.u.; exact two-port with losses and charging. The lossless $U^2/((1-k)\cdot X_L)$ would give 1540 MW uncompensated and 3080 MW at $k = 0.5$ — $R = 27.5\,\Omega$ and the charging are not negligible at this length.

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Voltage profile at heavy load



- 900 MW at cos $\varphi = 0.95$ lagging delivered at the receiving end, U_R held at 1.0 p.u.
- Uncompensated the sending end must sit at 1.217 p.u. — far outside the operating band.
- The midpoint bank lifts the voltage by $I \cdot X_C$ in the direction of power flow: U_S falls to 1.157, 1.127 and 1.107 p.u. for $k = 0.3$, 0.5 and 0.7.
- The receiving half of the line is unchanged — the bank acts only on what is upstream of it, which is why end-of-line banks and split banks are also used.

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Bank rating and protection

The bank is rated for the planned continuous line current, not for the conductor thermal limit:

$$U_C = I_{rated} \cdot X_C, \qquad Q_C = 3 \cdot I_{rated}^2 \cdot X_C$$

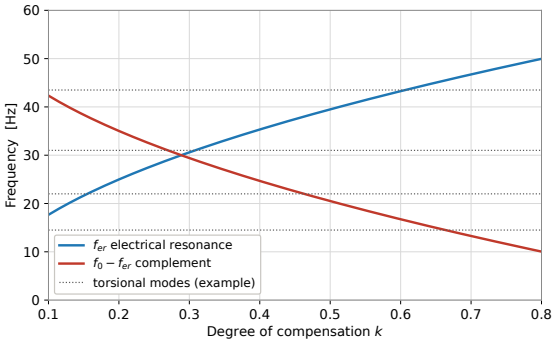
- Fault-current voltage is limited by a metal-oxide varistor (MOV) across the bank, protective level $2.0 \ldots 2.5 \cdot U_C$, with bypass breaker and damping circuit.
- Platform insulation at line potential; the bank carries the full fault current until the bypass closes.
- Emergency ratings (30 min, 10 s) are specified separately from the continuous rating.

Quantity		Value
k		0.5
X_C	Ω	81.2
C per phase	µF	32.7
I_{rated}	A	2000
U_C per phase	kV	162
Q_C three-phase	Mvar	974
MOV level ($2.3 \cdot U_C$)	kV	373

I_{rated} chosen as $\approx 1.15\times$ the current at 1500 MW, 500 kV. Midpoint bank on the reference line.

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Sub-synchronous resonance — screening



Torsional modes shown are illustrative; real values come from the generator manufacturer.

Series resonance of bank, line and source:

$$f_{er} = f_0 \cdot \sqrt{\frac{X_C}{X_L + X_S}}$$

- Rotor currents see the complement $f_0 - f_{er}$, which must stay clear of the turbine-generator torsional modes (10–45 Hz for large steam units).
- $X_S = 25\,\Omega$ (15 % on 1500 MVA): $k = 0.5$ gives $f_{er} = 39.5$ Hz, complement 20.5 Hz — within 2 Hz of the 22 Hz example mode.
- A hit at screening level calls for a detailed SSR study or a different k; type-3 wind adds SSCI.

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OHL Conductors

Definitions and applications

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OHL conductors — codewords

Codeword	Al/St strands
Cuckoo	24/7
Drake	26/7
Mallard	30/19
Condor	54/7

The four standard strandings of 795 kcmil ACSR.

- A size alone does not define a conductor: 795 kcmil ACSR exists in four standard strandings, each with a different steel content and so different strength, weight and sag.
- The codeword fixes area, stranding and core in one word.
- Originally Alcoa product nomenclature (1920s–30s), later adopted into the ASTM standards (ACSR: ASTM B232).
- The name family identifies the construction:
  - ACSR / ACSS — birds (Drake, Falcon, Rail, Cardinal)
  - AAC — flowers (Rose, Iris, Tulip, Peony)
  - AAAC — US city names (Akron, Amherst, Anaheim)
- No size information in the name: Bluebird is 2156 kcmil, Condor only 795 kcmil.

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OHL conductors — codewords elsewhere

- BS 215 and Indian practice use the same idea with mammals: Dog, Wolf, Lynx, Panther, Zebra, Moose. A 400 kV Indian line is specified as “twin Moose”, the North American equivalent as “bundled Drake”.
- Metric designs in continental Europe usually drop codewords and quote the aluminium and steel areas directly, e.g. 490/65 mm^2.
- Practical consequence for us: a conductor must always be quoted with its standard as well as its codeword, since the same name can appear in more than one national system.

ASTM B232 (ACSR), ASTM B231 (AAC), ASTM B399 (AAAC), BS 215-2

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Link collection

Overhead line (OHL) conductor data

- southwire.com/wire-cable/bare-aluminum-overhead-transmission-distribution

AC cable data

- new.abb.com — XLPE submarine cable systems (2GM5007)

PSCAD tutorials and information

- pscad.com/knowledge-base/topic-281
- pscad.com/knowledge-base/article/44

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AC High Voltage Equipment

Design limitations

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Current calculation

- Note that for current calculation the phase-phase voltage V_{LL} is used
- The maximum current is limited not only by the DC line capability but also by equipment on the DC line such as circuit breakers. The maximum current rating for circuit breakers is $5000A$ (Hitachi Energy, GE).

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